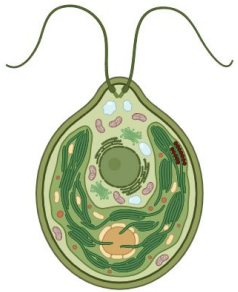


Gastrointestinal Tract Drug Delivery Using Algae Motors

Blake Estefan, Ruiming Tang, Qingqing Wang, Ishaan Mehrotra, Noah Campbell

BME577



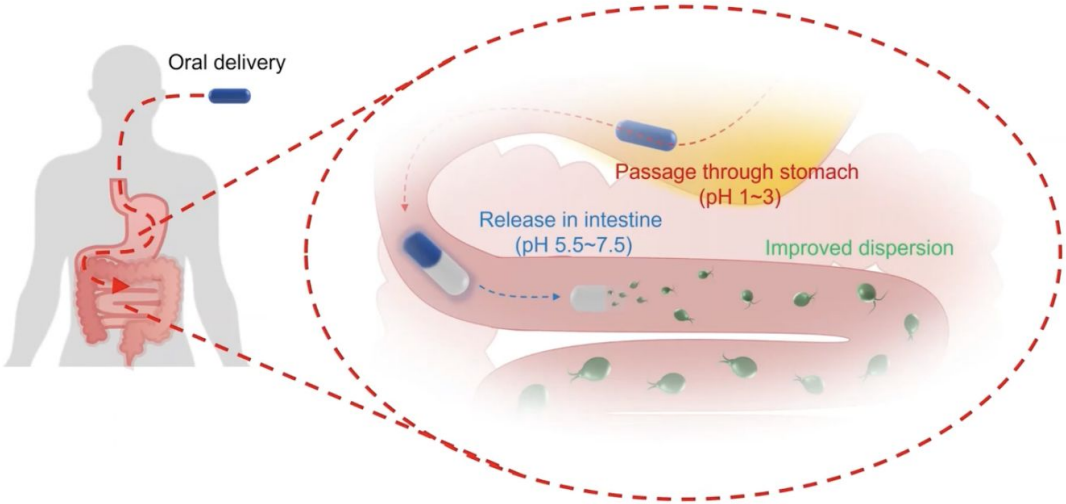
Background & Introduction

Why Improving GI Drug Delivery Matters ?

- Oral drug delivery is convenient, noninvasive, and patient-friendly.

<u>Key challenges:</u>

- Harsh stomach acid
- Short interaction with intestinal wall
- Rapidly cleared before full absorption

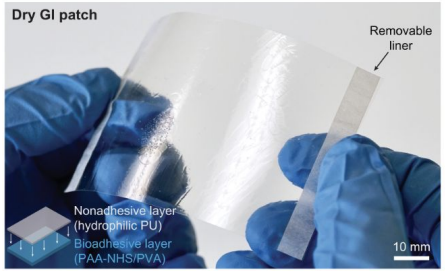


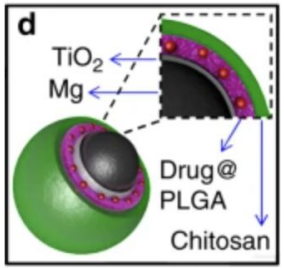
Fangyu Zhang *et al.* ,Gastrointestinal tract drug delivery using algae motors embedded in a degradable capsule.*Sci. Robot.***7**,eabo4160(2022).DOI:10.1126/scirobotics.abo4160

Previous Techniques to Enhance GI Delivery

Strategy	Advantages	Limitations
Microneedles	Penetrate mucosa	Invasive, device complexity
Hydrogels & Patches	Improve retention	Passive, short release
Mg-based micromotors	Active motion	Propulsion lasts < 15 min

- Despite progress, no current system achieves long-lasting active propulsion in the intestine with biocompatibility and safe delivery.



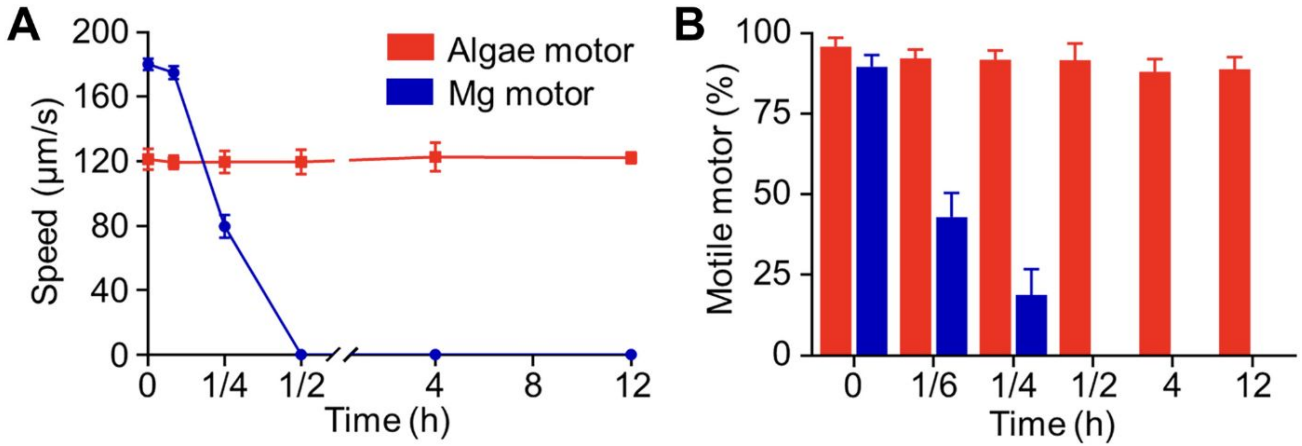


Abramson, A., Caffarel-Salvador, E., Soares, V. *et al.* A luminal unfolding microneedle injector for oral delivery of macromolecules. *Nat Med* 25, 1512–1518 (2019).

Jingjing Wu *et al.* ,An off-the-shelf bioadhesive patch for sutureless repair of gastrointestinal defects.*Sci. Transl. Med.***14**,eabh2857(2022).

de Ávila, B.EF., Angsantikul, P., Li, J. *et al.* Micromotor-enabled active drug delivery for in vivo treatment of stomach infection. *Nat Commun* 8, 272 (2017).

Challenge: Short Lifespan of Synthetic Micromotors



- Synthetic micromotors (e.g., Mg) lose activity quickly in GI fluids.
- Cannot sustain motion long enough to reach target tissue.

Can we find a motor system that's safe, easy to produce, and capable of long-lasting motion in the GI tract?

Algae-Based Micromotor: A Natural Solution

Uses *Chlamydomonas reinhardtii* as biohybrid motor

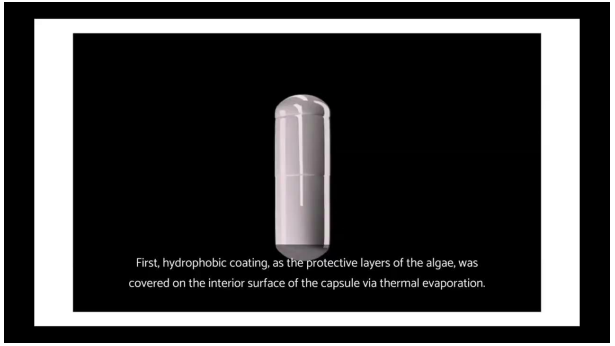
Benefits:

- l2 hours propulsion in intestinal fluid
- Biocompatible, scalable, and autofluorescent
- Protected by a dual-layer capsule (inner hydrophobic + outer pH-sensitive)

➢ First ever study to leverage **self-propulsion** of live algae for oral delivery

How it works?

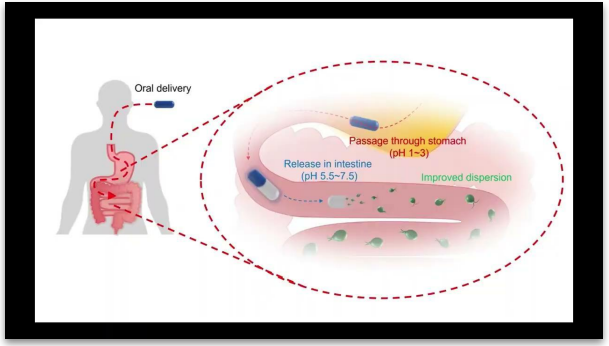
Algae Motor–Loaded Capsule System



1. **Algae Carrier (** Core component) :
   - Green algae (Chlamydomonas reinhardtii) serve as natural, self-propelled carriers.
   - Chemical modification attaches drug-loaded nanoparticles.

2. **Protective Capsule**
   - **Inner Coating** : Inner hydrophobic organosilicon (OTMS) layer retains moisture and algae viability;
   - **Outer Coating** : outer Eudragit Ll00-55 coating remains stable in the stomach and dissolves in the intestine, releasing the algae micromotors along with their drug payload.

Algae Motor–Loaded Capsule System



After oral administration :

1. Capsule first enters the stomach, where the enteric coating protects it from degradation at acidic gastric pH.
2. Upon entering the intestines, the coatings are dissolved in the nearly neutral pH and the capsule is degraded, leading to complete release of the algae motors.
3. Flagella-driven algae maintain prolonged mucosal contact, enhancing drug absorption, reducing clearance, and boosting local concentration for improved efficacy.

Comparison of **Motion Properties** Between **Algae-Based** and **Mg** Micromotors

—— In Vitro Simulated Intestinal Fluid (SIF)

Algae-Based

Mg

Movie S1

Movie S2

In SIF, algae motors maintained a constant speed of 120 μm/s for at least 12 hours, while Mg micromotors halted completely within 30 minutes. This underscores the superior long-term, rapid self-propulsion of algae motors in SIF compared to Mg micromotors.

Comparison of **Motion Properties** Between **Algae-Based** and **Mg** Micromotors

—— In Vivo Mouse Evaluation

A **B**

The algae motors displayed a broad intestinal distribution compared to the narrow signal from Mg motors.



It confirmed that the sustained propulsion of algae results in improved retention

The observed differences in biodistribution suggested that algae, with their long-lasting movement properties, could be effective at delivering drug payloads locally within the GI tract.

In Vitro and In Vivo Evaluation of an Algae Motor–Loaded Capsule System

Release of algae motors in a capsule in vitro

A



B



C

- The algae motors could be released from the modified capsule within a short period.

GI tract delivery of algae motors in comparison with other algae controls

- Free algae produced almost no signal, underscoring the necessity of capsule protection against gastric acid.
- Capsule-delivered algae motors exhibited broader distribution and enhanced retention in the intestines, highlighting the importance of self-propulsion.

In Vitro and In Vivo Evaluation of an Algae Motor–Loaded Capsule System

Feasibility of Algae Motor-Driven GI Therapeutic Delivery

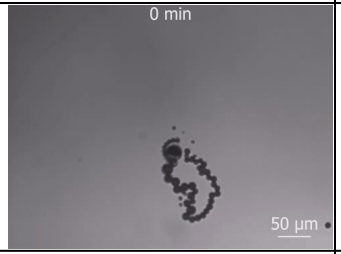
A B C

1. Unchanged Release and Cytotoxicity:
Binding NP(Dox) to algae motors **does not alter the drug release profile** , and **the cytotoxic efficacy** against melanoma cells is **maintained** .

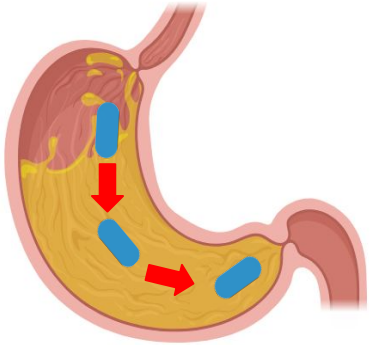
2. Enhanced In Vivo Delivery:
Capsules containing algae-NP(Dox) deliver **significantly higher drug levels** to the small intestine than NP(Dox) alone, demonstrating **improved retention and therapeutic potential** .

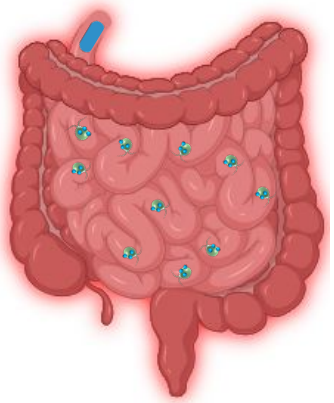
Pros

Prolonged propulsion

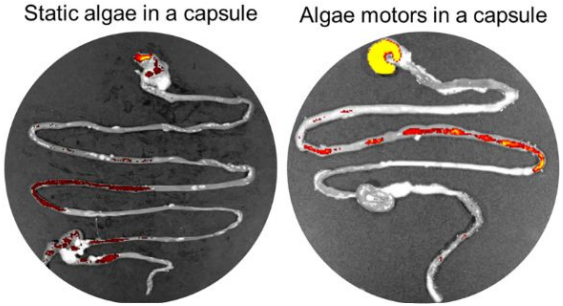


Targeted delivery



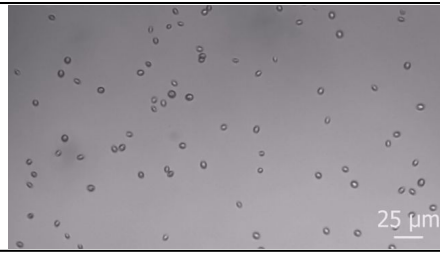


Enhanced retention

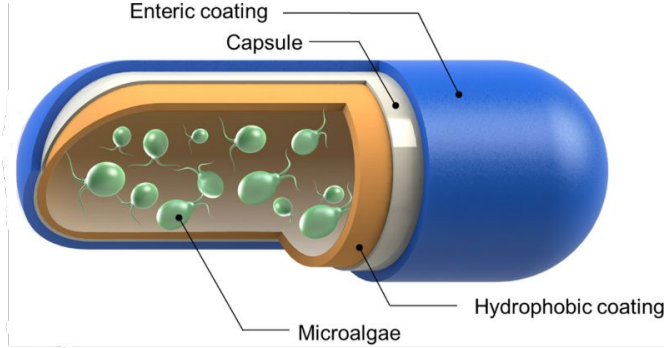


Cons

Limited real-time control over algae directionality

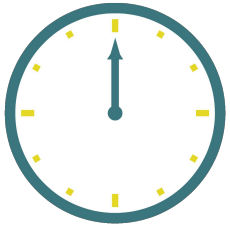


Complex manufacturing / scalability challenges



Variation in degradation timing



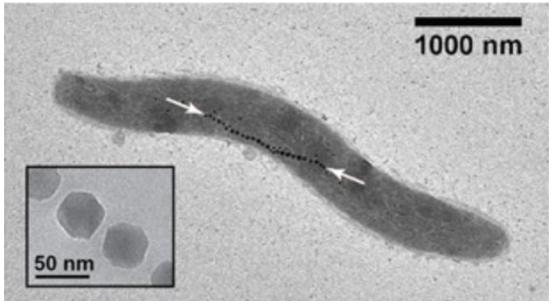





0 min
50 µm

Movie S8

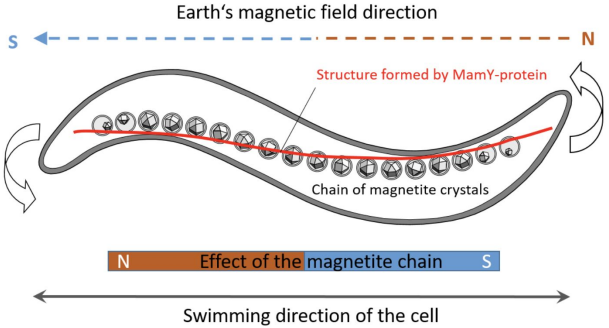
Novel Improvement (Part 1)



Ideas for improvement:

- Use a different microorganism motor system
  - C. reinhardtii (used in this paper) is quite large (10um[1] vs 20um intestinal cell size)
  - Alternative: Magnetospirillum magneticum
    - 2.4-5.7um in length and <1um in width[2]
    - Still well functionalized for bioconjugation of drugs

- Importantly:
  - M. magneticum are still propelled by bodily movement, but can also be guided by magnetic fields.
  - Further, research has shown a preference for low oxygen environments (such solid tumor or other deep tissue), and efficient internalization by mammalian cells.[3]

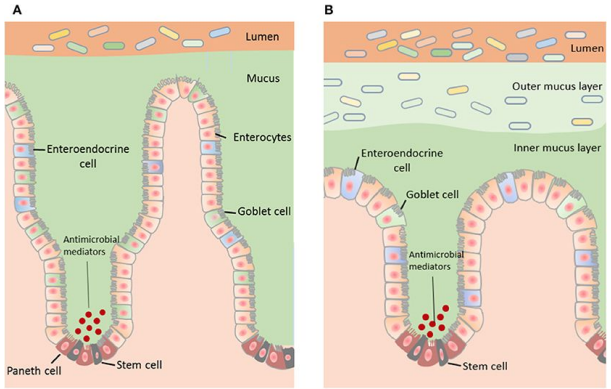


[1] Harris, E. (2001). Chlamydomonas as a model organism. Annu Rev Plant Physiol Plant Mol Biol. 2001 Jun52: 363-406. p.368
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[3] Gandia, D., et al., (2022). ACS Applied Materials & Interfaces, 15(1), 566–577. https://doi.org/10.1021/acsami.2c18435

Novel Improvement (Part 2)

Ideas for improvement:

- Genetically encoded enzymes to degrade mucus
  - The intestinal mucus lining significantly reduces the diffusivity of drugs into target cells in the intestinal crypts.[1]
  - While research is limited, some have proposed Enzyme Decorated Drug Carriers [2], which could locally deliver enzymes capable of permeating the intestinal glycocalyx.
- Genetically encoded targeting/adhesion molecules
  - Micromotors edited to express proteins that would enhance cellular attachment would allow for better – potentially even targeted– tissue retention. Such bacteria have already been engineered, & some tested in clinical trials for cancer therapy.[3]

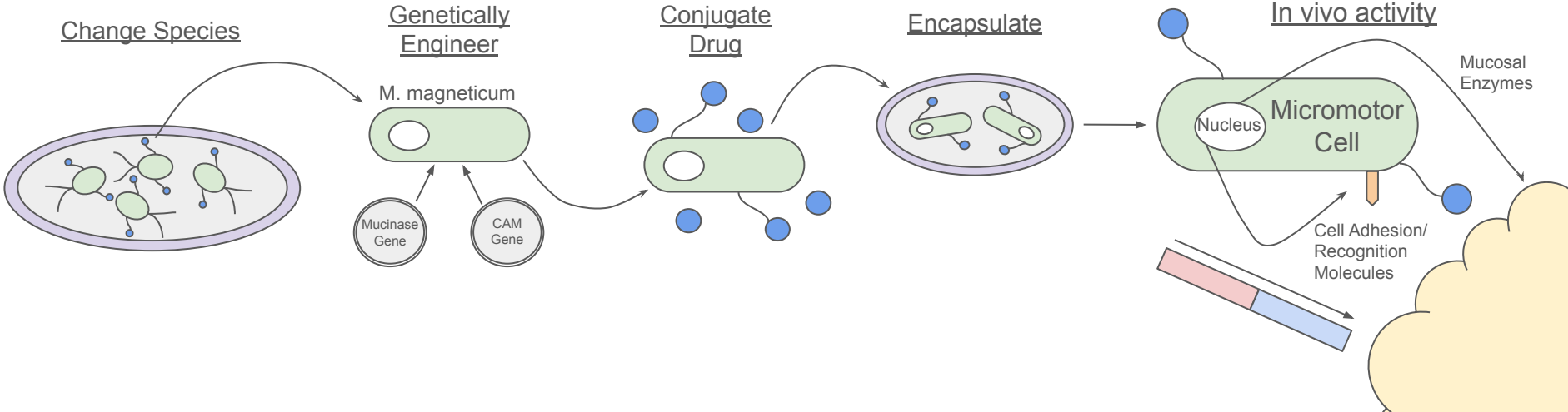


[1] Falavigna M, Stein PC, Flaten GE, di Cagno MP. Pharmaceutics. 2020 Feb 17;12(2):168. doi: 10.3390/pharmaceutics12020168.
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[3] Gurbatri, C. R., Arpaia, N., & Danino, T. (2022). Science, 378(6622), 858–864. https://doi.org/10.1126/science.add9667

Composed Novelty: Magnetotactic Microorganisms for Enhanced GI Drug Delivery

Combine both for transgene expression (mucosal enzymes or cell adhesion molecules) in smaller and more “directable” micromotor organisms.

Initial Model → Improved Model



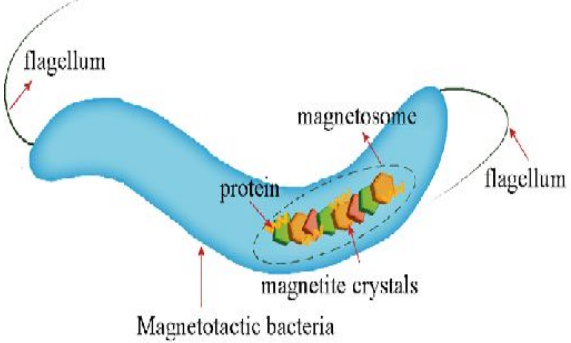
Improvement (Novelty)

- **Replace large algae** (10–20 μm) with **smaller magnetotactic bacteria** that carry internal magnetite chains and transgenes for **improved permeability and retention** in intestinal tissue.

- **Benefit:** Combine **self-propulsion (flagella)** with **magnetic guidance** for more precise tissue targeting and more effective drug delivery by minimizing undesired clearance.

- **Still encapsulated** using the same pH-sensitive, degradable capsules from the original paper for **safe delivery** through the stomach.

Impact & Significance → Global Health



- **Better targeting → Higher Efficacy**
  - Magnetotaxis + smaller size = improved navigation through tight or hypoxic regions (e.g. tumors)
  - Reduces off-target drug exposure, making treatments safer and potentially more effective.

- **Potential for Resource-Limited Settings**
  - Once bacteria are standardized, oral capsule delivery is patient-friendly, possibly lowering infrastructure needs (e.g. fewer clinic visits vs. IV therapy)

Impact & Significance → Economic Considerations

- **Reduced Drug Waste**
  - More precise targeting often means lower doses of expensive therapeutics (e.g. chemotherapeutics)

- **Cost-Benefit Balance**
  - Manufacturing magnetotactic bacteria at scale and formulating capsules may be costlier initially
  - Over time, lower side-effect profile and improved patient compliance could offset production costs

Impact & Significance → Environmental Benefits

- **Biodegradable and Bio-Based**
  - Like algae, these bacteria are naturally occurring, and the capsule materials degrade safely.

- **Decreased Pharmaceutical Pollution**
  - Enhanced targeting can reduce the total amount of drug released into the ecosystem.

Impact & Significance → Societal Impact

- **Less Invasive, Improved Compliance**
  - Oral micromotor capsules are more comfortable vs. invasive infusions or injections

- **Fewer Side Effects = Greater Acceptance**
  - Targeted delivery can minimize systemic toxicity, fostering broader public support

- **Future Therapeutic Frontiers**
  - Potential to treat hard-to-reach tumors and chronic GI diseases more effectively, improving quality of life

Impact & Significance → Potential Drawbacks

- **Manufacturing Scalability**
  - Culturing engineered magnetotactic bacteria in large quantities might require specialized bioreactors
- **Regulatory Hurdles**
  - Engineering living microorganisms for therapy will need thorough safety evaluations
- **Public Perception**
  - Societal acceptance of genetically (or otherwise) engineered microbes must be considered

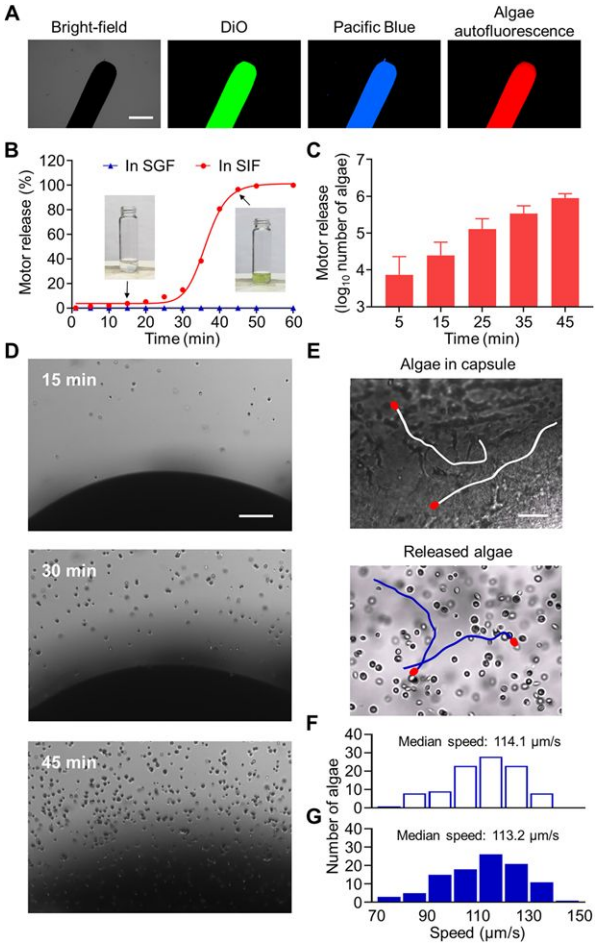
Thank You!

Questions?

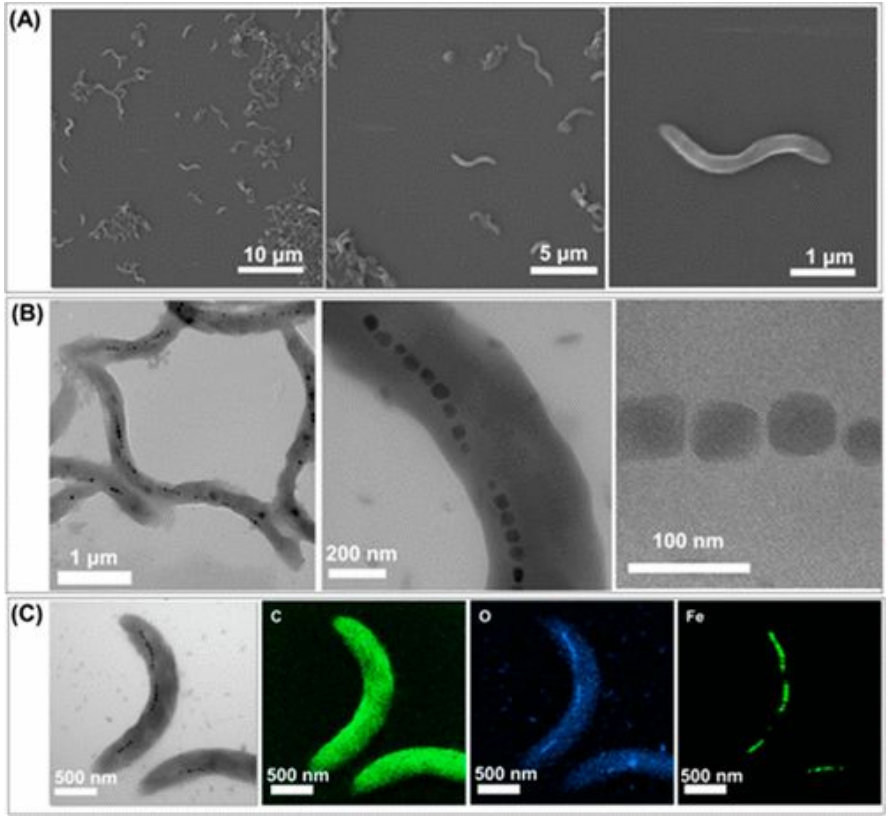
References

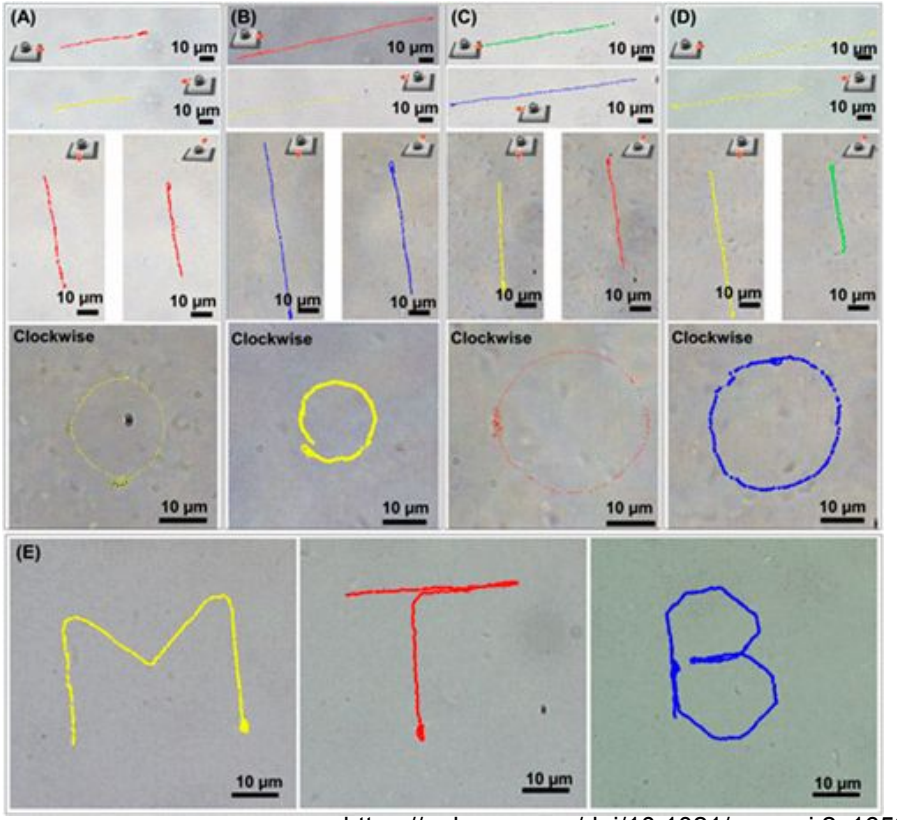
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Supplemental Information



Directionality Control





https://pubs.acs.org/doi/10.1021/acsami.2c16592